

(I) True/False (20')

1. (☒) It is possible to have an overlap between high and low levels in digital logic.
2. (☒) Full-adders cannot be used as a BCD-to-binary converter.
3. (☒) The XNOR gate will produce a high output if only one but not both of the input is HIGH.
4. (☒) NAND gates can be used to construct NOR gates.
- ★ 5. (☒) The effect of an inverted output being connected to the inverting input of another gate is to effectively eliminate one of the inversions, resulting in a single inversion.
6. (☒) The BCD equivalent of 73 is 01001001.
7. (☒) A D flip-flop is constructed by connecting an inverter between the Set and Clock terminals.
8. (☒) A J-K flip-flop associated waveforms are shown in Fig1-1, the circuit is operating properly

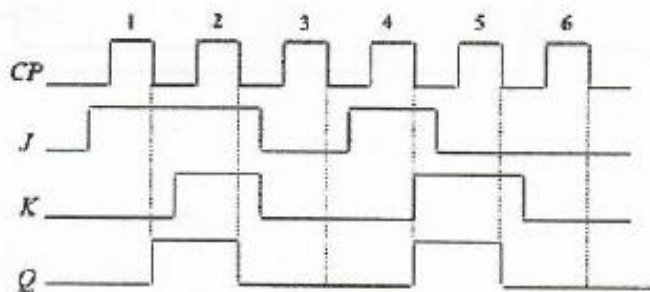


Fig 1-1

9. (☐) A 4-bit staircase-ramp A/D converter has a clock frequency of 1000Khz and maximum input voltage 5V. The minimum sample rate will be 6250 samples/second.
10. (☐) A 8-bit D/A converter has a resolution of 0.0125.

(二). Choice

1. Which of the following is correct for full-adders ()
- (A) Full-adders have the capability of directly adding decimal numbers
 - (B) In a parallel full-adder, the first stage may be a half-adder
 - (C) Full-adders are limited to two inputs, since there are only two binary digits
 - (D) Full-adders are used to make half-adders

2. Using 4 bit adders to create a 32 bit adder ()
- (A) requires 32 adders
 - (B) requires 8 adders
 - (C) requires the carry out of the less significant adder to be connected to the carry in of the next significant adder
 - (D) both (B) and (C)

3. What type of device is shown in Fig 2-1, and what inputs (A3, A2, A1, A0) are required to produce the output levels as shown ()

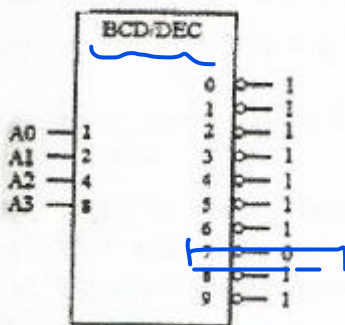


Fig 2-1

- (A) A decimal to binary-coded-decimal encoder; 1, 1, 1, 0
- (B) A binary-to-decimal encoder; 0, 1, 1, 1
- (C) A binary-coded-decimal to decimal decoder; 1, 0, 0, 0
- (D) A decimal to binary-coded-decimal encoder; 0, 0, 0, 1

(E) other

E 选项为 others

4. For the input values (A0-A3, B0-B3) given for Fig 2-2, what will be indicated on the display? 74139 is a 1-of-4 decoder ()
- (A) Data Select=1, B=6; Data Select=0, A=3
- (B) Data Select=1, A=6; Data Select=0, B=3
- (C) Data Select=1, A=3; Data Select=0, B=6
- (D) Data Select=1, B=3; Data Select=0, A=6

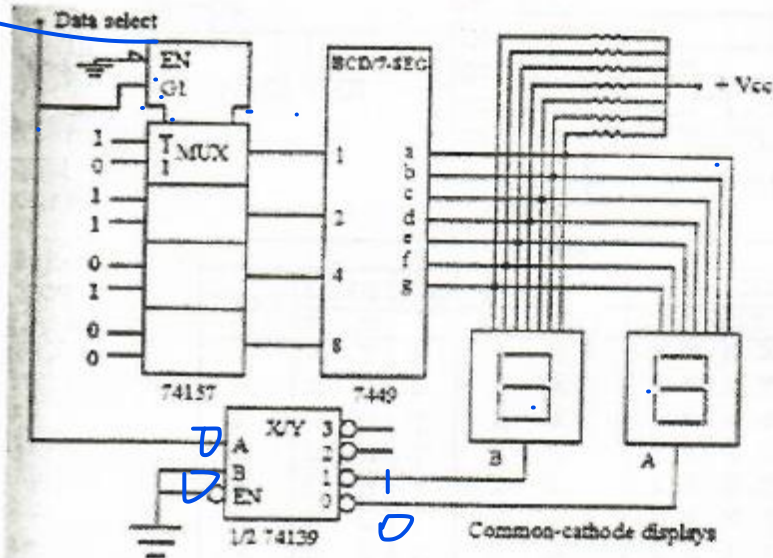


Fig 2-2

5. What is the frequency of the F_{out} in Fig 2-3 ()
- (A) 800Khz
- (B) 20Khz
- (C) 200Khz
- (D) 10Khz

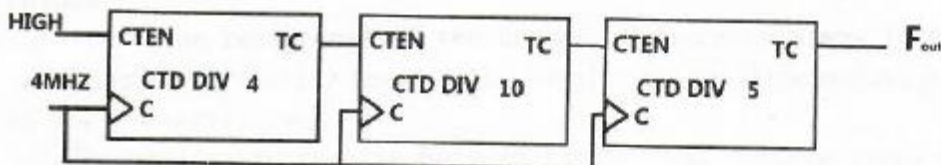


Fig 2-3

6. Which of the example below express the distributive law? ()
- (A) $A(B+C)=AB+AC$
- (B) $(A+B)+C=A+(B+C)$
- (C) $A+(B+C)=AB+AC$
- (D) $A(BC)=(AB)+C$

7. The circuit shown in Fig 2-4 is a/an ().
- (A) 4-bit magnitude comparator
(B) 4-bit half adder
(C) 8-bit comparator
(D) 8-bit binary adder

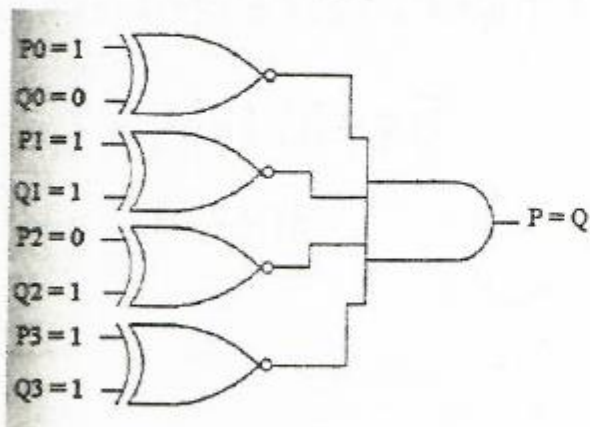


Fig 2-4

8. The state diagram of counter is Fig 2-5, The capacity of the counter is ().

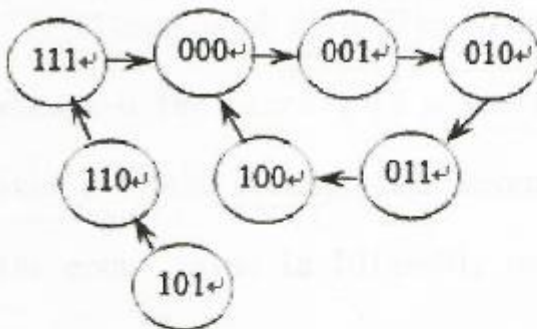


Fig 2-5

- (A) 4 (B) 5 (C) 6 (D) 8

9. Referring to the function table in Fig 2-6, taking the CLEAR, S1 and S0 inputs all HIGH will do what? ().

- (A) The exact response will depend on what values are loaded into the parallel data inputs
(B) It will cause the parallel data inputs to be loaded and passed to the parallel data outputs
(C) It will reset the parallel registers and inhibit the serial data inputs
(D) It will inhibit the operation of the register

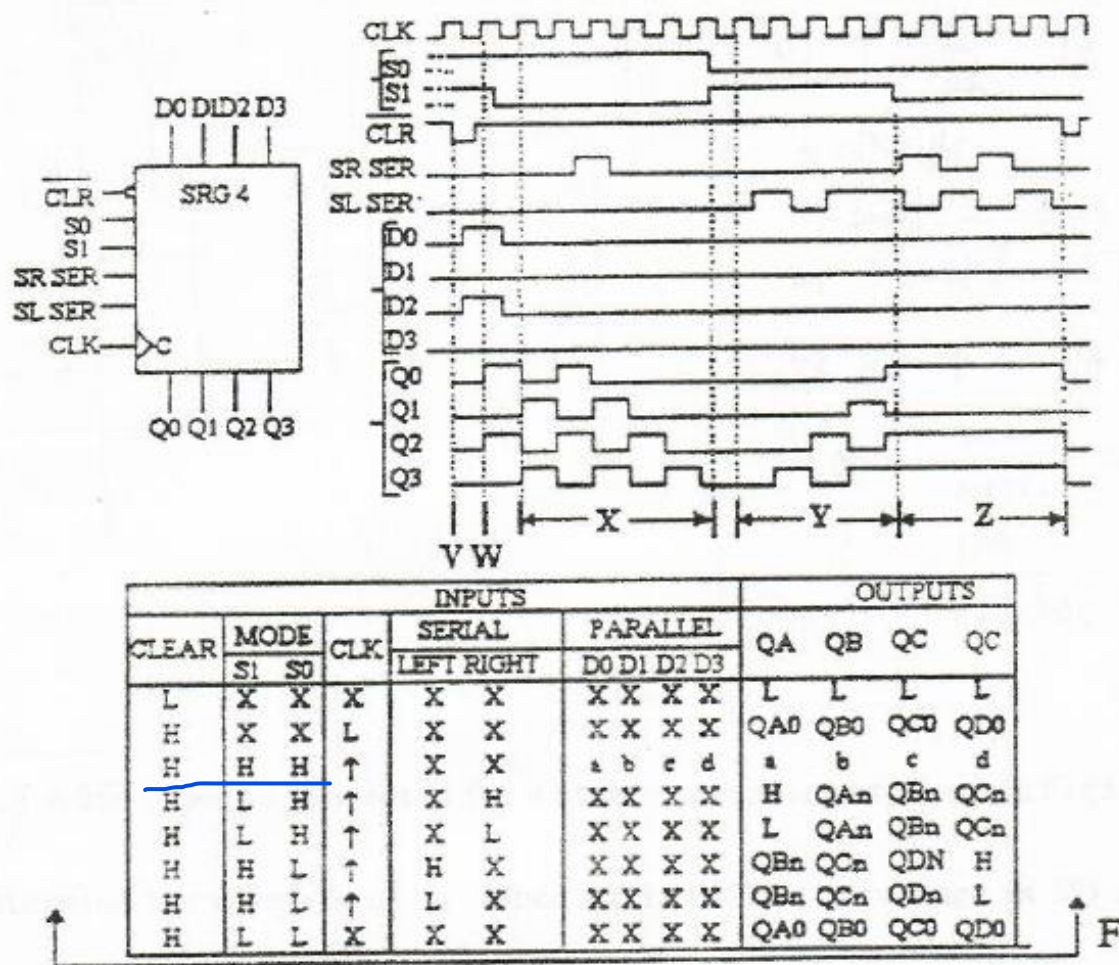


Fig 2-6

10. What is the accuracy of D/A converter? ()

- (A) It is the comparison between the actual output of the converter and its expected output
- (B) It is the reciprocal of the number of discrete steps in the D/A output
- (C) It is the deviation between the ideal straight-line output and the actual output of the converter
- (D) The ability to resolve between forward and reserve steps when sequenced over its entire range of inputs

(三) Problems

—) Using Algebra method to simplify the function. (10')

$$F = XZ + \bar{Y}Z + Y\bar{Q} + Z\bar{Q}C + Z\bar{Q}\bar{C} + X(Y + \bar{Z}) + \bar{X}YZ\bar{Q} + X\bar{Y}QC$$

二) Please use D flip-flop to design a synchronization timing circuit, when $x=0$, the circuit is a sextuple denotation counter (6 进制计数器); when $x=1$, the circuit is a Seven binary counter (7 进制计数器), and when the count value is $101^5(x=0)$, or $110(x=1)$, the output carry is $Z=1$.

三) A 555 timer is connected for a stable operation as shown in Fig3-1, Determine the waveform of v_o (When v_o is HIGH, its voltage is 5V) and v_c . The resistance of the bulb is 5 Ohm $R_1=R_3=10K$ ohm. IF I want the power of bulb is $3\sim 3.75W$. Please give the value of R_2 and the frequency of v_o . (10')

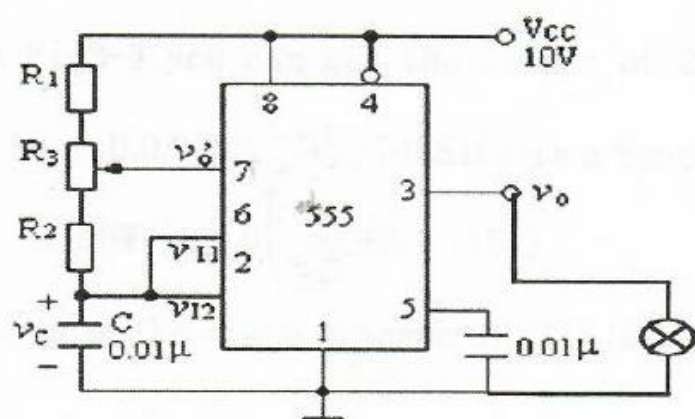
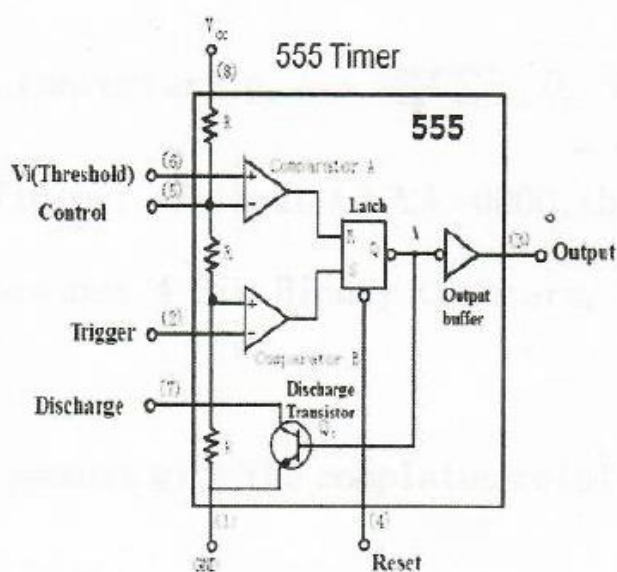


Fig 3-1



四) Analyse the circuit, determine the output waveform. (15')

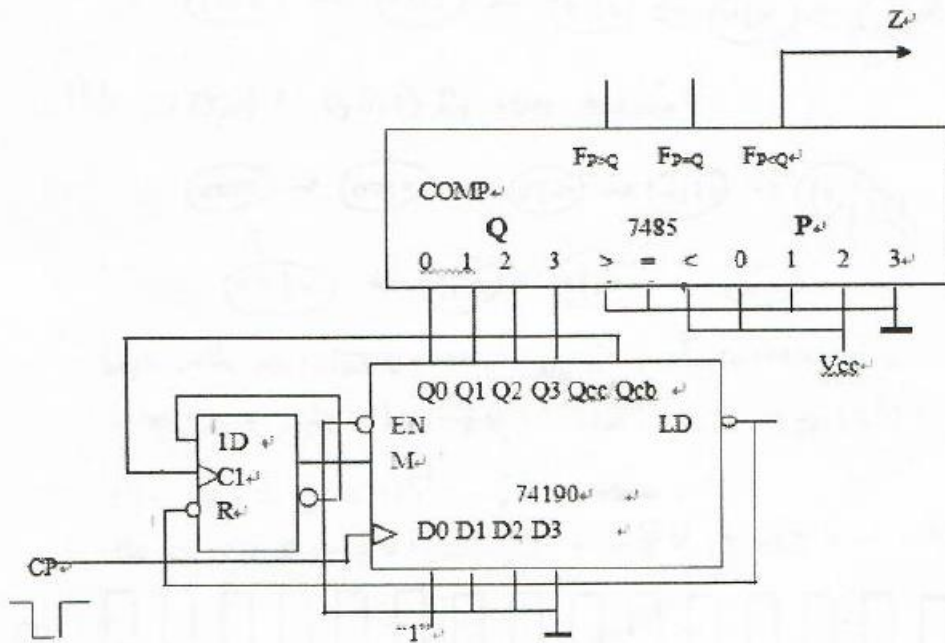


Fig 3-2

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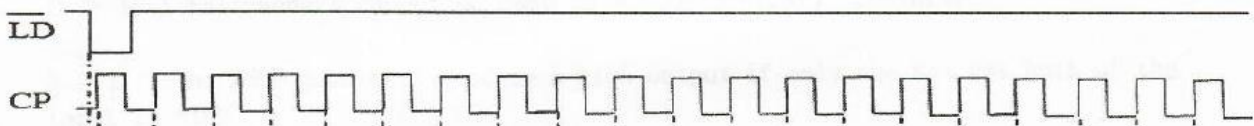
$\overline{EN}$	$\overline{LD}$	M	CP	D0	D1	D2	D3	Q0	Q1	Q2	Q3
x	0	x	x	a	b	c	d	a	b	c	d
0	1	0	↑	x	x	x	x	递增计数			
0	1	1	↑	x	x	x	x	递减计数			
1	1	x	x	x	x	x	x	保持			

$$Q_{cc}/Q_{cb} = \overline{M} Q_3 \overline{Q_2} \overline{Q_1} \overline{Q_0} + M Q_3 \overline{Q_2} \overline{Q_1} \overline{Q_0}$$

Q3—Most Significant Bit ; Q0—Least Significant Bit

7485

A3	B3	A2	B2	A1	B1	A0	B0	A>B	A=B	A<B	$F_{p>Q}$	$F_{p=Q}$	$F_{p<Q}$
0	1	x	x	x	x	x	x	x	x	x	0	0	1
1	0	x	x	x	x	x	x	x	x	x	1	0	0
A3=B3	0	1	x	x	x	x	x	x	x	x	0	0	1
A3=B3	1	0	x	x	x	x	x	x	x	x	1	0	0
A3=B3	A2=B2	0	1	x	x	x	x	x	x	x	0	0	1
A3=B3	A2=B2	1	0	x	x	x	x	x	x	x	1	0	0
A3=B3	A2=B2	A1=B1	0	1	x	x	x	x	x	x	0	0	1
A3=B3	A2=B2	A1=B1	1	0	x	x	x	x	x	x	1	0	0
A3=B3	A2=B2	A1=B1	A0=B0	1	0	0	0	1	0	0	1	0	0
A3=B3	A2=B2	A1=B1	A0=B0	0	1	0	0	0	1	0	0	1	0
A3=B3	A2=B2	A1=B1	A0=B0	0	0	1	0	0	0	1	0	0	1



波形图包含 Q0、Q1、Q2、Q3、Qcc/QcB、M、Z

五) In Fig3-3 , AD7524 is a 8-bit D/A converter, $\mu_o = -\frac{V_{REF}}{2^8} \sum_{i=0}^7 D_i \cdot 2^i$.

in Fig3-4 you can get the output of 2716(eg: if input $A_3A_2A_1A_0=0000$; then output: $D_3D_2D_1D_0=0000$), 74LS161 is a Synchronous 4-Bit Binary Counters, IF L_D is LOW , $Q_DQ_CQ_BQ_A=DCBA$. (15')

(1) Draw the state diagram of 74LS161, you must give the completes results

(2) Determine the μ_o waveform for CP (This signal is clock)

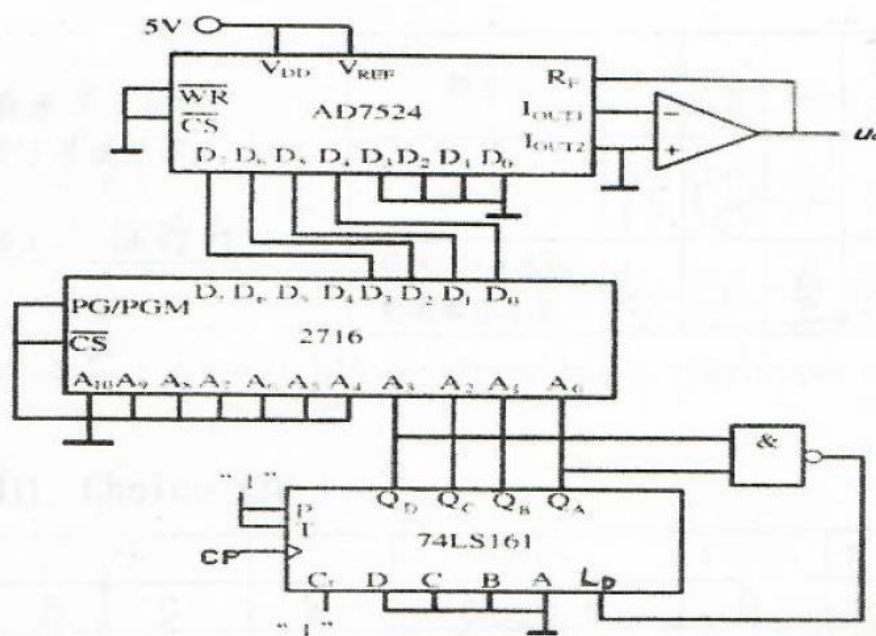


Fig 3-3

地 址				数 据			
A_3	A_2	A_1	A_0	D_3	D_2	D_1	D_0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	1	0	0	1	0	0
0	0	1	1	0	1	1	0
0	1	0	0	1	0	0	0
0	1	0	1	1	0	0	0
0	1	1	0	0	1	1	0
0	1	1	1	0	1	0	0
1	0	0	0	0	0	1	0
1	0	0	1	0	0	0	0
1	0	1	0	1	0	0	0
1	0	1	1	0	1	1	0
1	1	0	0	0	0	1	0
1	1	0	1	0	1	0	0
1	1	1	0	0	0	1	0
1	1	1	1	0	1	0	0

Fig 3-4